



# Principal Component Analysis



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## High-Dimensional Data

Most traditional statistical techniques for regression and classification are intended for the low-dimensional setting in which  $n$  is much greater than  $p$ .

Let's consider predicting the blood pressure based on age, gender and body mass index (BMI). There are three or four (if an intercept is included) predictors in the model and thousands of patients with the predictor values. Hence  $n \gg p$  and the problem is low-dimensional.

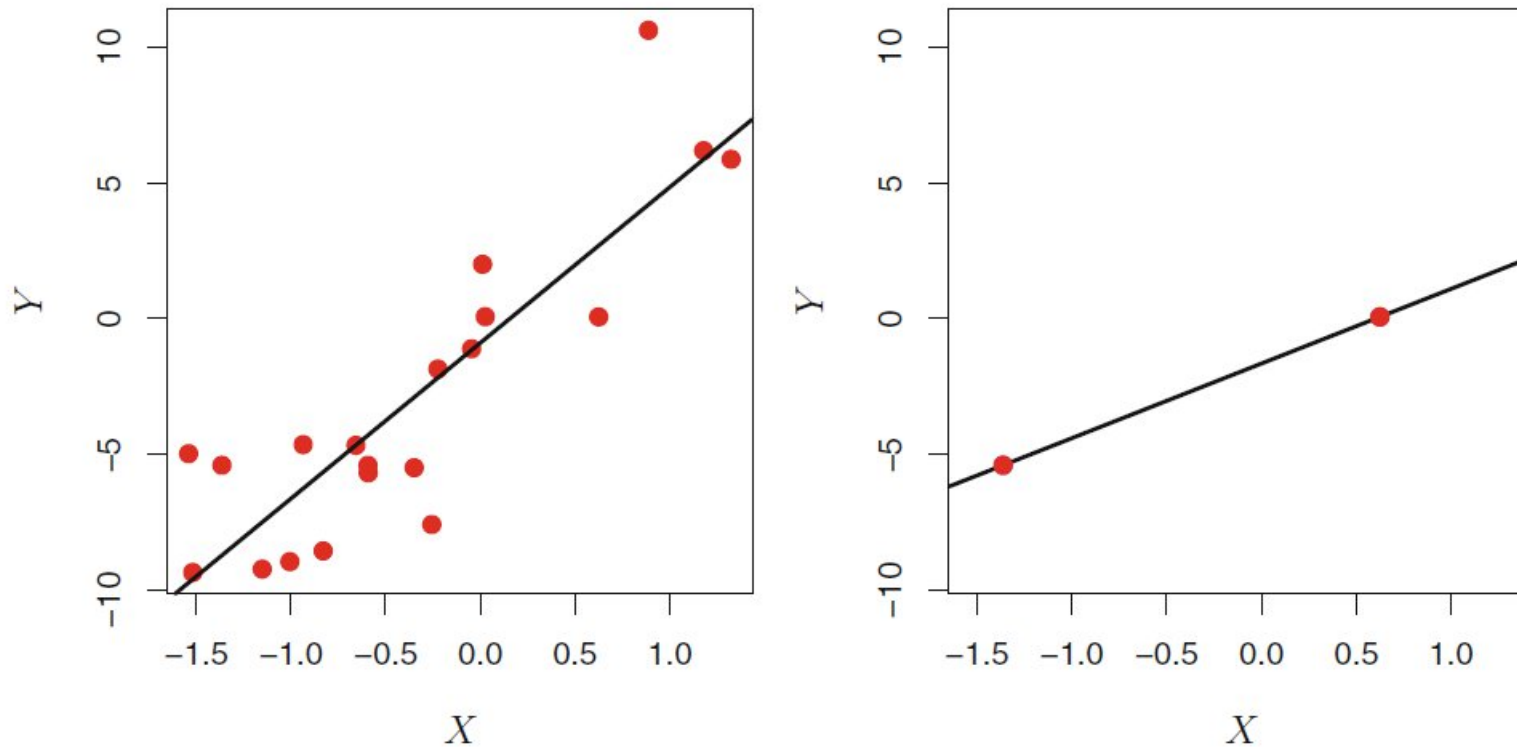
But now it is common to collect a large number of feature measurements (in fields such as Finance, marketing and medicine). While  $p$  can be extremely large  $n$  is often limited due to cost, sample availability or other considerations.

Datasets containing more features than observations are called as High-dimensional. Classical approaches such as least squares linear regression are not appropriate in this case.

## What Goes Wrong in High Dimensions?

When  $p$  is as large as or larger than  $n$ , least squares cannot or should not be performed. Because regardless of whether or not there is a relationship between the features and the response, least squares will yield a set of coefficient estimates that result in a perfect fit to the data such that the residuals are zero.

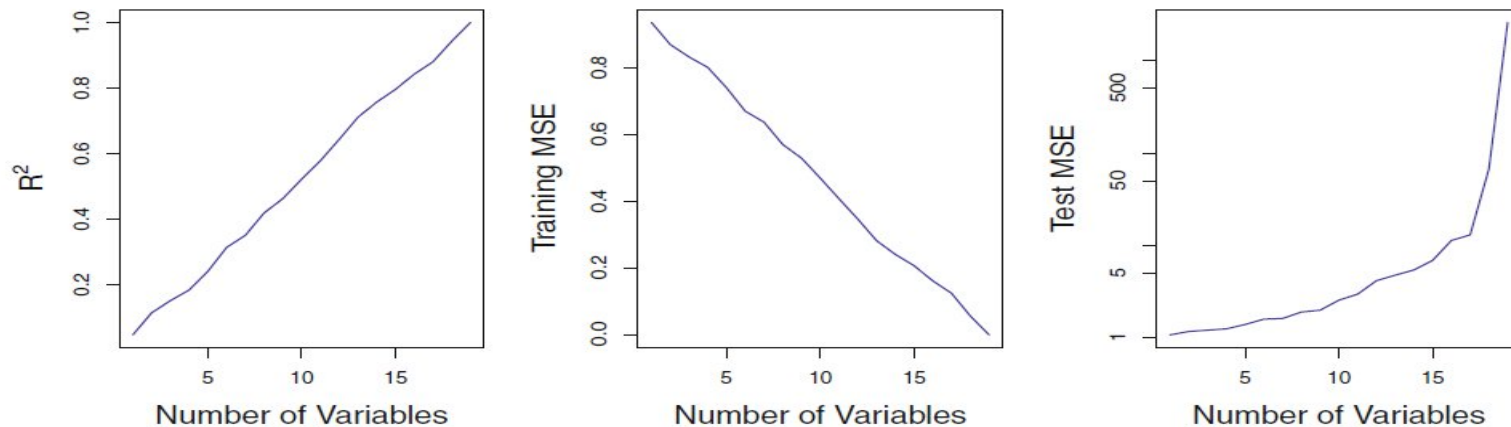
Consider following example.



## What Goes Wrong in High Dimensions?

When there are 20 observations  $n > p$  and the least squares regression line does not perfectly fit the data. Instead, the regression line approximate the 20 observations as well as possible. When there are only two observations, regardless of the values of those observations the regression line will fit the data exactly. This is problematic because this perfect fit will almost certainly lead to over fitting. Though it is possible to perfectly fit the training data in the high-dimensional setting, the resulting model will perform extremely poorly on test data.

When  $p > n$  or  $p$  is close to  $n$ , a simple least squares line is too flexible and hence overfits the data. Following plots shows the risk of high dimensions when the number of variables is increased.



## What Goes Wrong in High Dimensions?

Another problem in high-dimensional data is the Curse of Dimensionality. This is the situation, as the number of features or dimensions grows, the amount of data we need to generalize accurately grows exponentially.

In a high-dimensional feature space with each feature having a range of possible values, typically an enormous amount of training data is required to ensure that there are several samples with each combination of values. A typical rule of thumb is that there should be at least 5 training examples for each dimension in the representation.

Consider the following example.

Probably a kid will like to eat cookies, so let us assume that you have a whole truck with cookies having a different colour, a different shape, a different taste, a different price ...



## What Goes Wrong in High Dimensions?

If the kid has to choose but only take into account one characteristic e.g. the taste, then it has four possibilities: sweet, salt, sour, bitter, so the kid only has to try four cookies to find what (s)he likes most.

If the kid likes combinations of taste and colour, and there are 4 different colours, then he already has to choose among  $4 \times 4$  different types;

If he wants, in addition, to take into account the shape of the cookies and there are 5 different shapes then he will have to try  $4 \times 4 \times 5 = 80$  cookies

We could go on, but after eating all these cookies he might already have belly-ache ... before he can make his best choice :-) Apart from the belly-ache, it can get really difficult to remember the differences in the taste of each cookie.

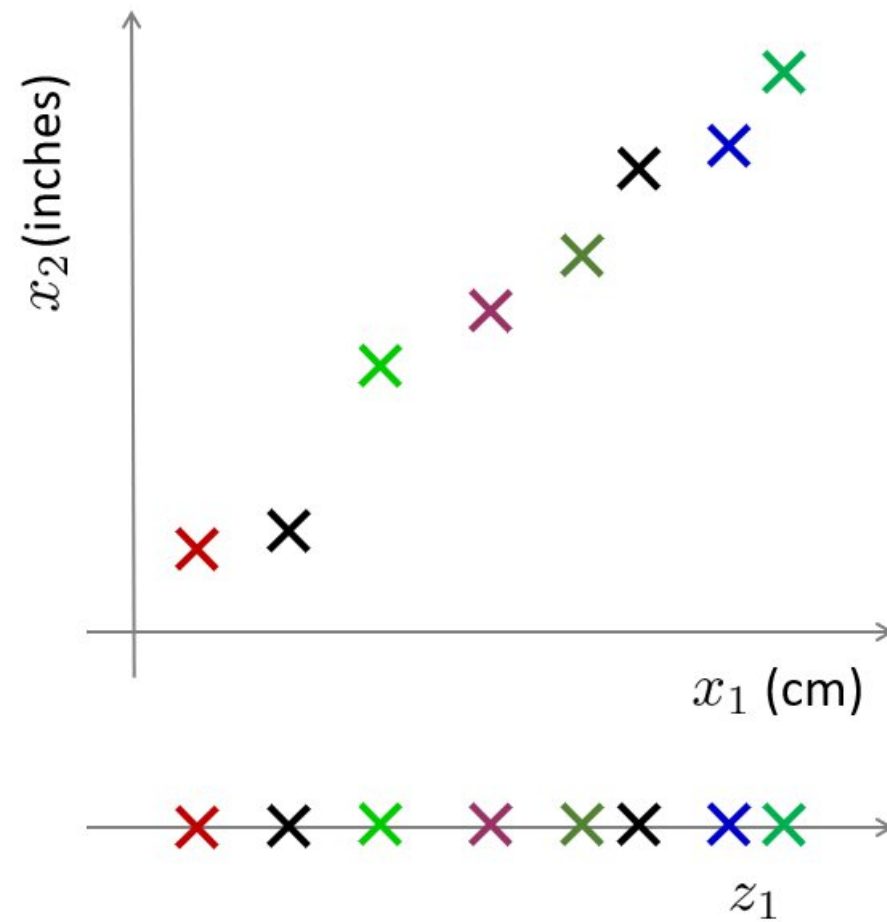
## Dimensionality reduction

Simplify complex high-dimensional data. Summarize data with a lower dimensional data.

- Given data points in  $d$  dimensions
- Convert them to data points in  $r < d$  dimensions
- With minimal loss of information

## Dimensionality reduction

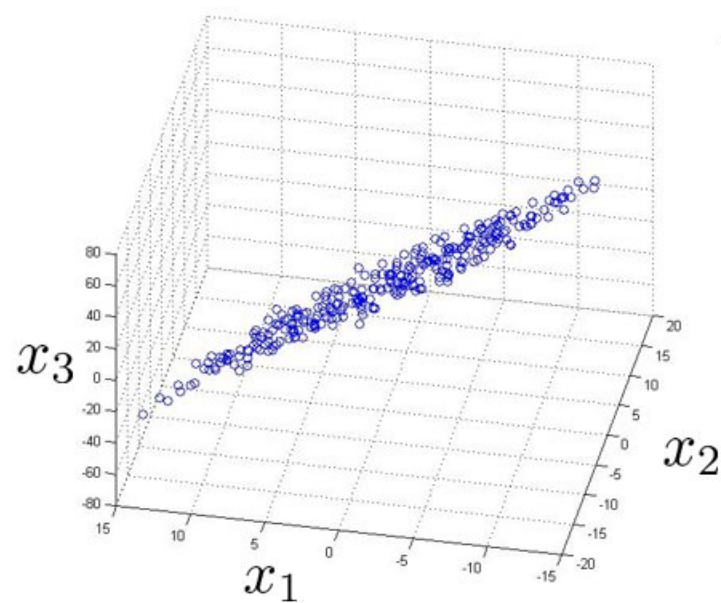
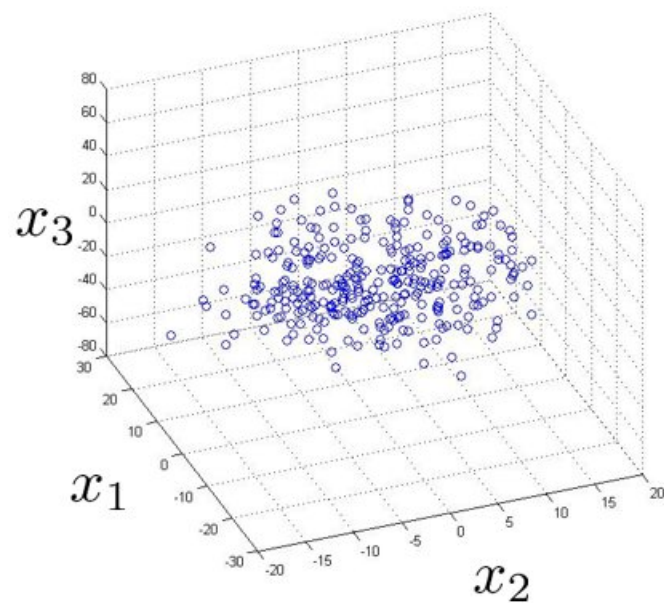
Reduce data from 2D to 1D





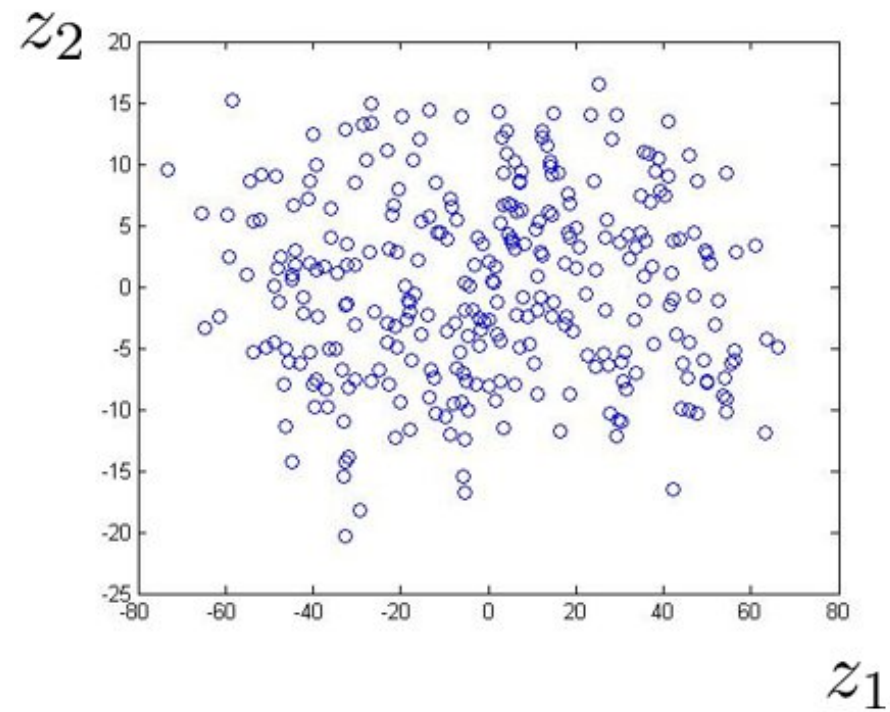
## Dimensionality reduction

Reduce data from 3D to 2D



## Dimensionality reduction

Reduce data from 3D to 2D



## **Dimensionality reduction**

There are several ways to do dimensionality reduction.

1. Principal Component Analysis
2. Factor Analysis
3. Partial Least Square Regression
4. Feature Selection
5. Regularization (LASSO, Ridge, Elastic-Net)
6. Independent Component Analysis
7. Multidimensional Scaling

And there are many more. In this discussion we are going to discuss Principal Component Analysis (PCA).

## Principal Component Analysis

Principal components allow us to summarize a large set of correlated variables with a smaller number of representative variables (Latent variables). These new variables collectively explain most of the variability in the original set.

Principal component analysis (PCA) is the process that compute principal components and the use of these components in understanding the data.

Each dimension found by PCA is a linear combination of the original  $p$  features.

## First Principal Component

The first principal component of a set of features  $X_1, X_2, \dots, X_p$ , is the normalized linear combination of the features which has the largest variance. So it can be written as,

$$Z_1 = \phi_{11}X_1 + \phi_{21}X_2 + \dots + \phi_{p1}X_p$$

Where,  $\sum_{j=1}^p \phi_{j1}^2 = 1$  (Unit Vectors)

The elements,  $\phi_{11}, \phi_{21}, \phi_{31}, \dots, \phi_{p1}$  are known as **loadings** of the first principal component.

So the loading vector can be written as,

$$\phi_1 = (\phi_{11} \ \phi_{21} \ \dots \ \phi_{p1})^T$$

## First Principal Component

How to compute the 1<sup>st</sup> principal component?

We assume that each of the variables in  $X$  are centered to have mean zero (the column means of  $X$  are zero) since we are only interested in variance.

We then look for the linear combination of the sample feature values of the form

$$z_{i1} = \phi_{11}x_{i1} + \phi_{21}x_{i2} + \cdots + \phi_{p1}x_{ip}$$

that has largest sample variance, subject to the constraint that,

$$\sum_{j=1}^p \phi_{j1}^2 = 1$$

## First Principal Component

In other words, the 1<sup>st</sup> principal component loading vector solves the optimization problem,

$$\underset{\phi_{11}, \dots, \phi_{p1}}{\text{maximize}} \left\{ \frac{1}{n} \sum_{i=1}^n \left( \sum_{j=1}^p \phi_{j1} x_{ij} \right)^2 \right\} \quad \text{subject to} \quad \sum_{j=1}^p \phi_{j1}^2 = 1$$

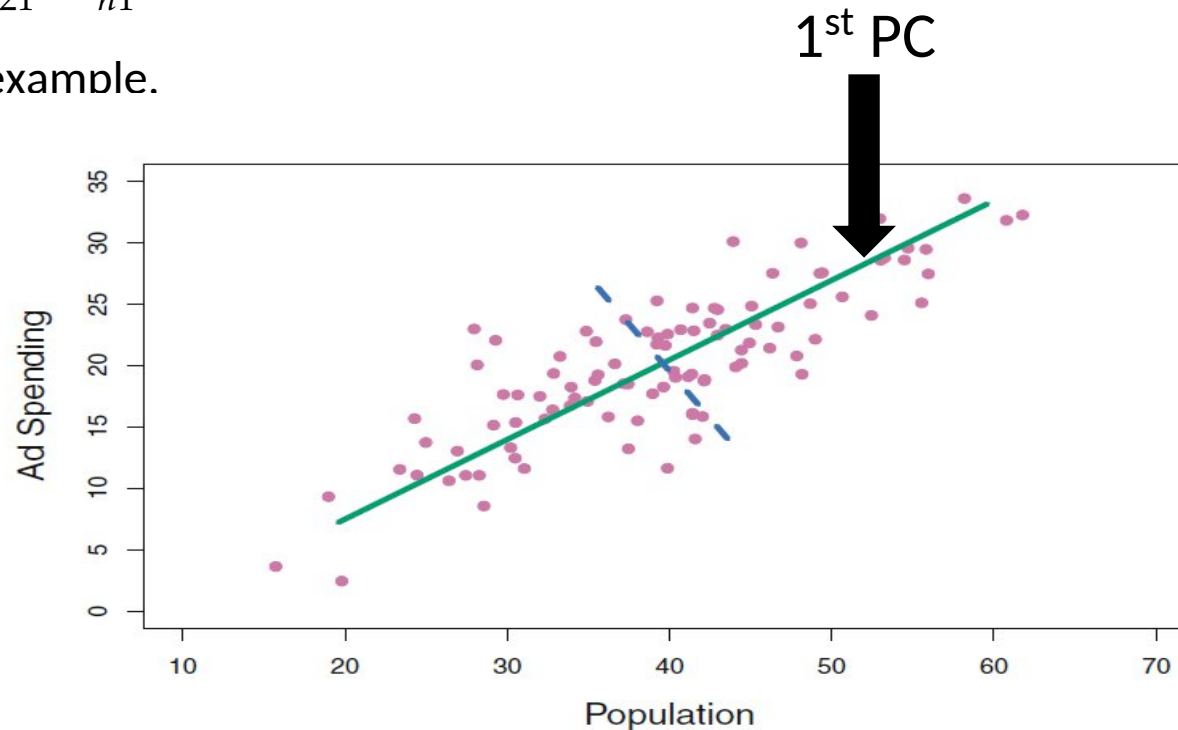
## First Principal Component

We refer to  $z_{11}, z_{21}, \dots, z_{n1}$  as the scores of the 1<sup>st</sup> principal component.

The loading vector  $\phi_1$  with elements  $\phi_{11}, \phi_{21}, \dots, \phi_{p1}$  defines a direction in feature space along which the data vary the most.

If we project the  $n$  data points  $x_1, x_2, \dots, x_n$  onto this direction, the projected values are the principal component scores  $z_{11}, z_{21}, \dots, z_{n1}$ .

Consider the following example.





## Second Principal Component

Once the 1<sup>st</sup> principal component  $Z_1$  is determined, we find the second principal component  $Z_2$ .

Second principal component is also a linear combination of  $X_1, X_2, \dots, X_p$  that has maximal variance out of all linear combinations that are uncorrelated with  $Z_1$ .

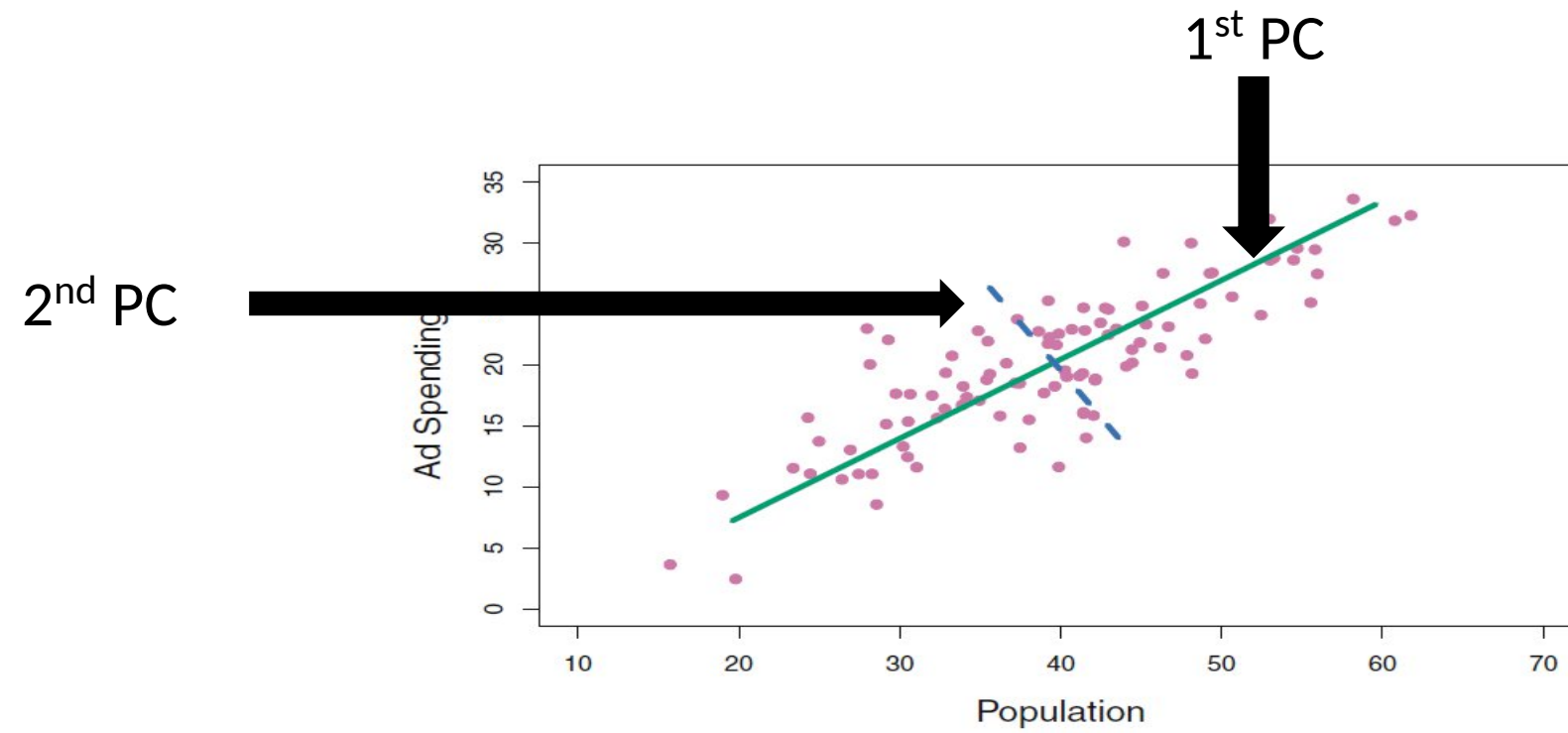
The second principal component scores  $z_{12}, z_{22}, \dots, z_{n2}$  take the form,

$$z_{i2} = \phi_{12}x_{i1} + \phi_{22}x_{i2} + \dots + \phi_{p2}x_{ip}$$

The elements,  $\phi_{12}, \phi_{22}, \phi_{32}, \dots, \phi_{p2}$  are known as **loadings** of the second principal component.

Second principal component is uncorrelated with the first principal component such that the loading vector of the second principal component is **orthogonal** to the loading vector of the first principal component.

## Second Principal Component



## Other Principal Components

So if  $p$  predictors are in the model,  $p$  principal components can be created.

$$Z_1 = \textit{Principal Component 01 (PC1)}$$

$$Z_2 = \textit{Principal Component 02 (PC2)}$$

$$Z_3 = \textit{Principal Component 03 (PC3)}$$

$$Z_4 = \textit{Principal Component 04 (PC4)}$$

$$Z_5 = \textit{Principal Component 05 (PC5)}$$

$\vdots$

$$Z_p = \textit{Principal Component } p \text{ (PC}p\text{)}$$

## Selecting Principal Components

But all of these principal components are not used. Few of these are selected such that the maximal variation of the data is captured.

We are interested in knowing the proportion of variance explained (PVE) by each principal component.

The total variance present in a dataset (assuming that the variables are centered to have mean zero) is defined as below.

$$\sum_{j=1}^p \text{Var}(X_j) = \sum_{j=1}^p \frac{1}{n} \sum_{i=1}^n x_{ij}^2$$

## Selecting Principal Components

Variance explained by the  $m^{th}$  principal component is given by,

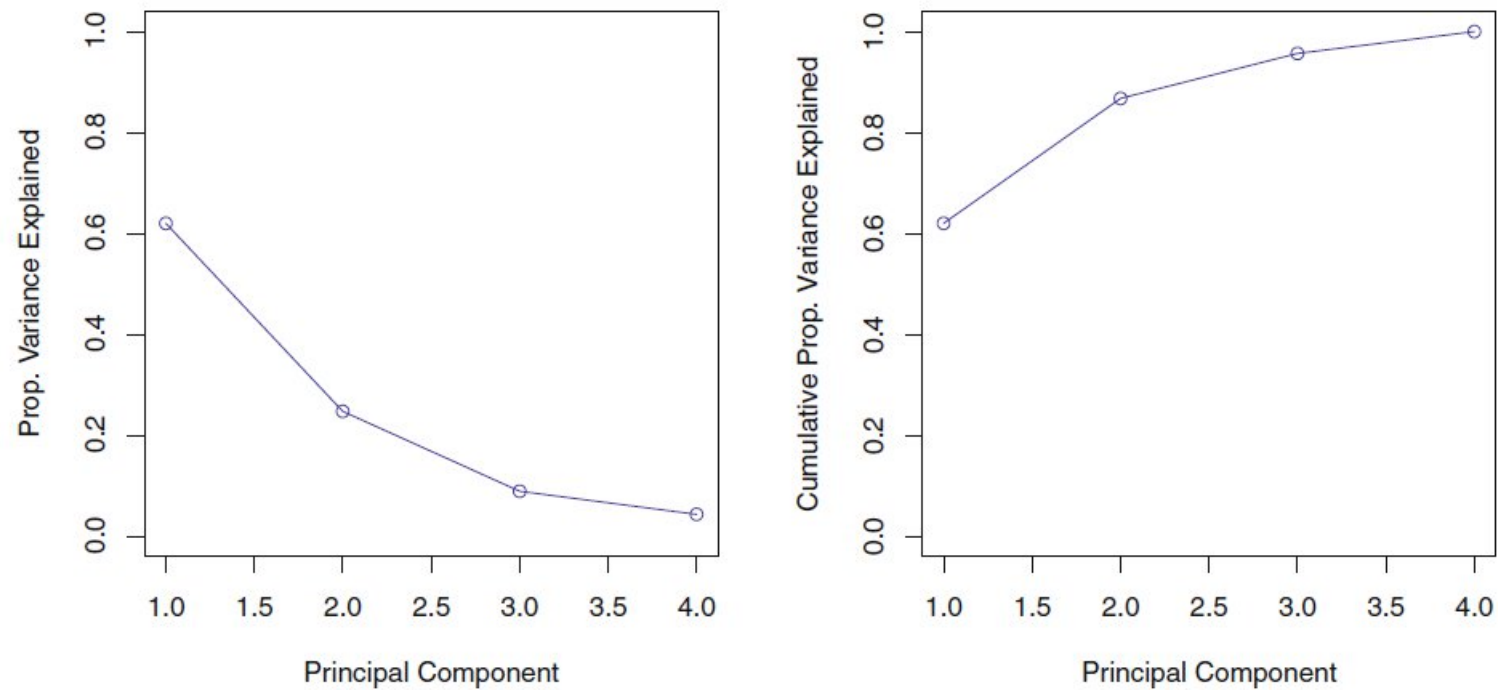
$$\frac{1}{n} \sum_{i=1}^n z_{im}^2 = \frac{1}{n} \sum_{i=1}^n \left( \sum_{j=1}^p \phi_{jm} x_{ij} \right)^2$$

PVE of the  $m^{th}$  principal component is given by,

$$\frac{\sum_{i=1}^n \left( \sum_{j=1}^p \phi_{jm} x_{ij} \right)^2}{\sum_{j=1}^p \sum_{i=1}^n x_{ij}^2}$$

## Selecting Principal Components

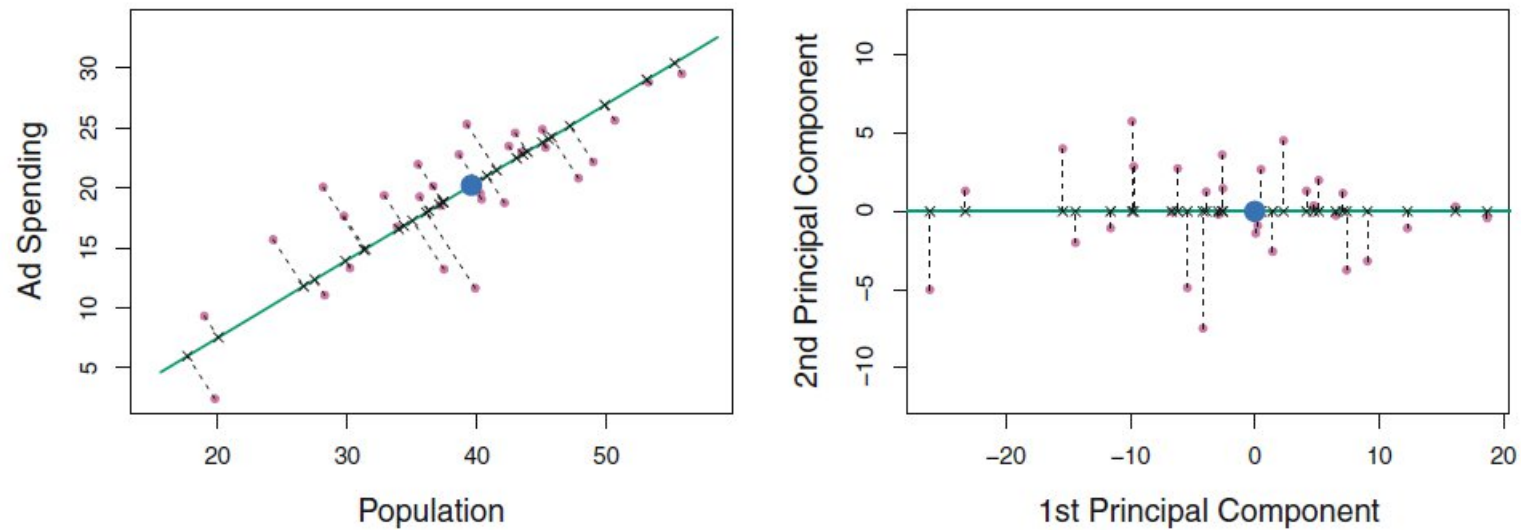
So PVE can be plotted against the number of principal components as below example,



↑  
Scree Plot

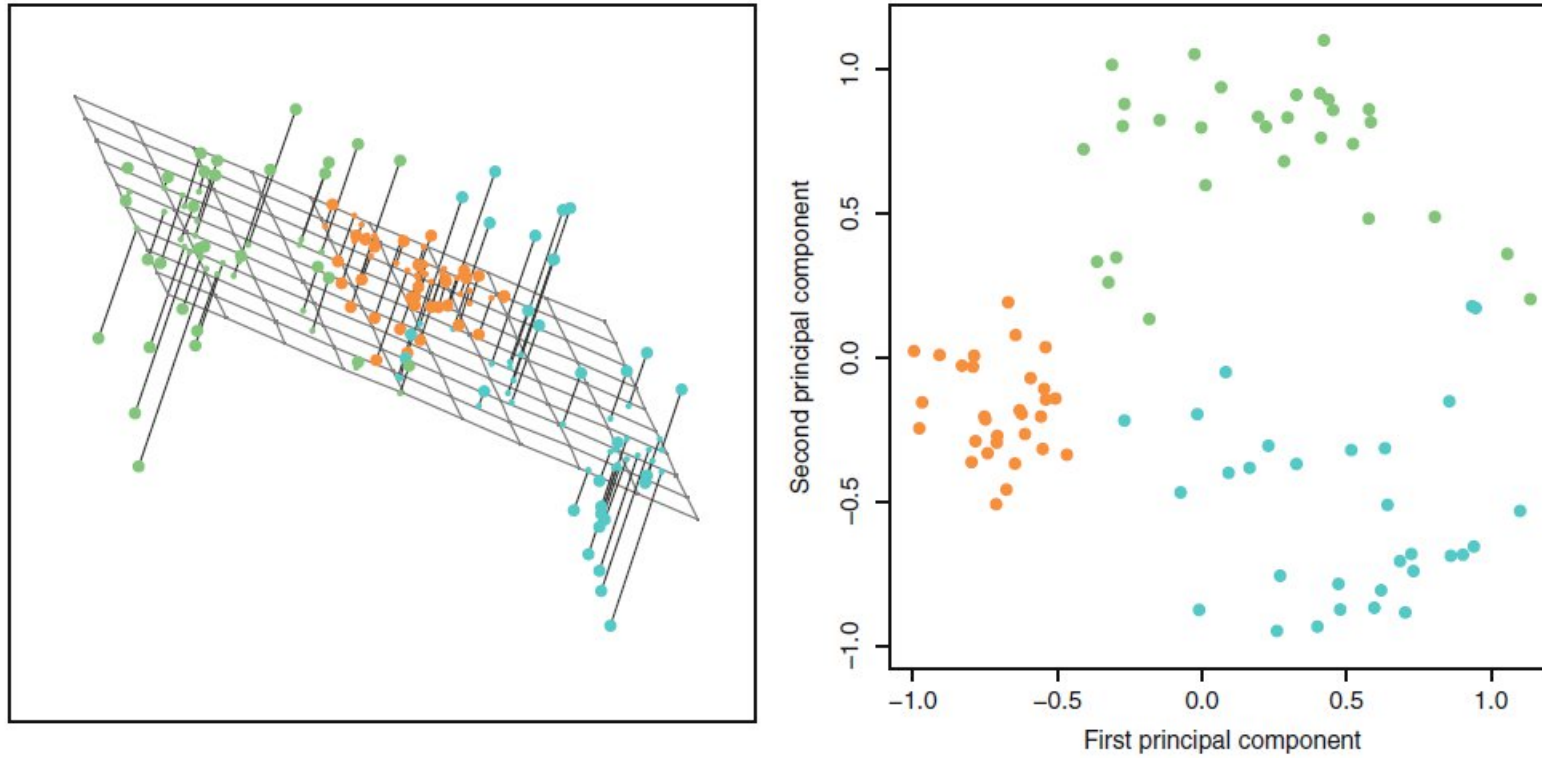
## Selecting Principal Components

From the PVE values, it can be identified that which set of PC s combinedly captures a satisfiable amount of the total variation of the data by reducing dimensions. Check the following examples



Two variables have been explained by a single new variable (PC1)

## Selecting Principal Components



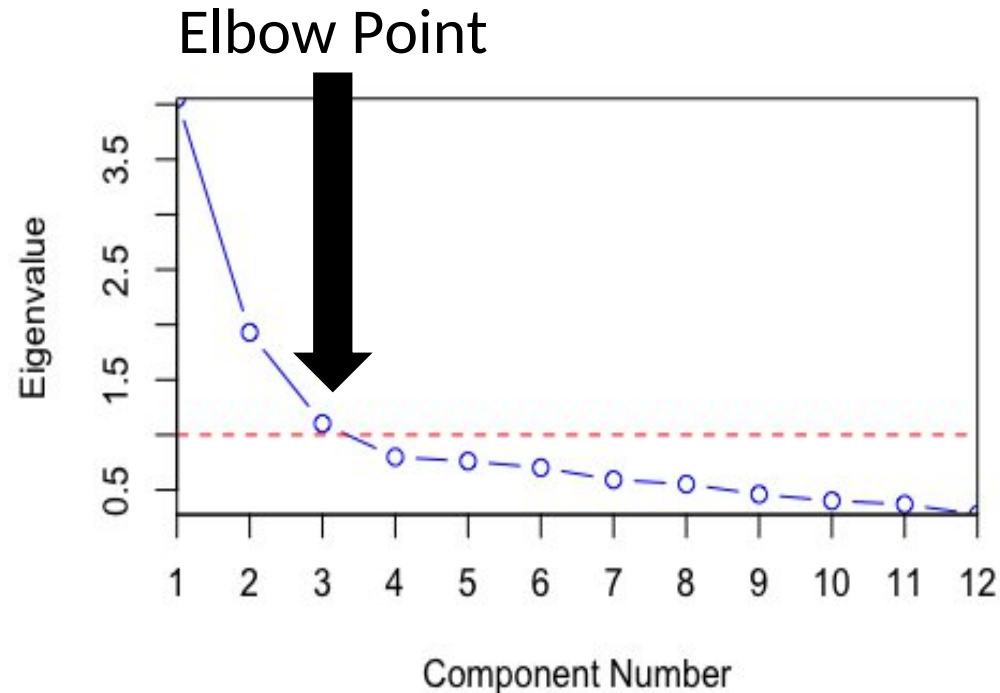
Three variables have been explained by two new variables (PC1 and PC2)



## Selecting Principal Components

How to select the principal components for representing the data and for reducing dimensions?

Actually these variances captured by each component are the **eigenvalues of the Variance - Co Variance matrix of variables**. Using them, we can select PC s which capture the majority of the variation using the scree plot. Consider the following scree plot as an example.



## Selecting Principal Components

How to select the principal components for representing the data and for reducing dimensions?

We typically decide the number of principal components required to visualize the data by examining a scree plot. We choose the smallest number of principal components needed to explain a sizable amount of the variation in the data. Done by looking for a point in the scree plot where the proportion of variance explained by each subsequent principal component drops off.

## Computing Principal Components

Consider

$$X = \begin{bmatrix} 2 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$$

When standardized

$$X_{\text{standardized}} = X - \text{mean} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \\ 0 & 0 \end{bmatrix}$$

Covariance matrix.

$$C = \frac{1}{N-1} X_{\text{standardized}}^T X_{\text{standardized}} = \frac{1}{2} \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \end{bmatrix}^T$$

$$C = \frac{1}{2} \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

## Computing Principal Components

Eigenvalues

$$\lambda_1 = 2, \lambda_2 = 0$$

Normalized Eigenvectors

$$\mathbf{u}_1 = \frac{1}{\sqrt{2}}[1, -1]^T$$

$$\mathbf{u}_2 = \frac{1}{\sqrt{2}}[1, 1]^T$$

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

## Computing Principal Components

Project each standardized data point  $x_i$  into principal component space

$$Z = X_{\text{standardized}} \cdot U$$
$$Z = \begin{bmatrix} 1 & -1 \\ -1 & 1 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} \sqrt{2} & 0 \\ -\sqrt{2} & 0 \\ 0 & 0 \end{bmatrix}$$

Final New Dataset in Principal Component Space

$$Z = \begin{bmatrix} \sqrt{2} & 0 \\ -\sqrt{2} & 0 \\ 0 & 0 \end{bmatrix}$$

## Applications Principal Components

Consider the following example.

For each of the 50 states in the United States, the dataset contains the number of arrests per 100,000 residents for each of three crimes: Assault, Murder, and Rape. We also record UrbanPop, the percent of the population in each state living in urban areas.

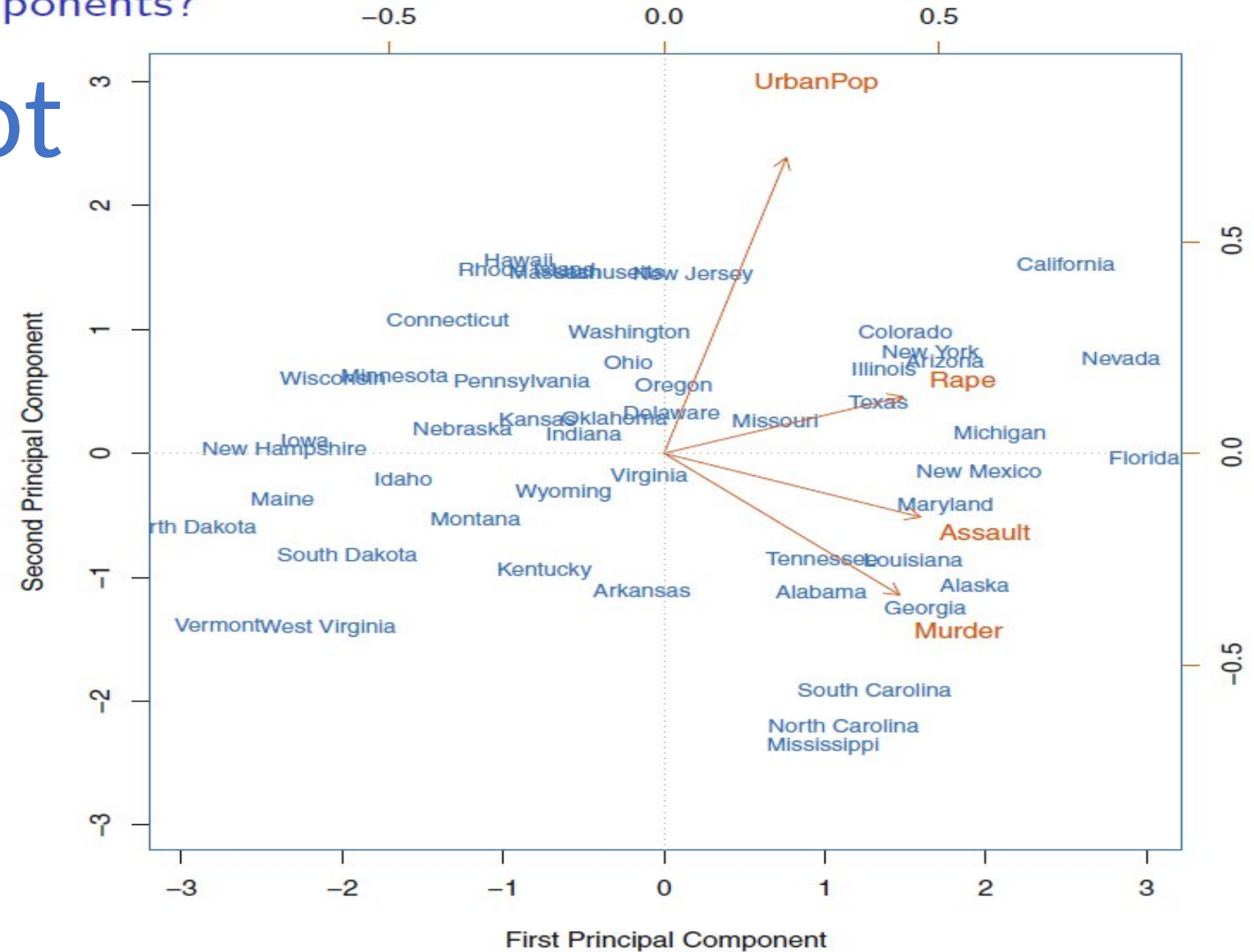
The principal component score vectors have length  $n = 50$  and the principal component loading vectors have length  $p = 4$ . PCA is performed after standardizing each variable to have mean zero and standard deviation one.

|          | PC1       | PC2        |
|----------|-----------|------------|
| Murder   | 0.5358995 | −0.4181809 |
| Assault  | 0.5831836 | −0.1879856 |
| UrbanPop | 0.2781909 | 0.8728062  |
| Rape     | 0.5434321 | 0.1673186  |

## Applications Principal Components

What Are Principal Components?

This is biplot



## Applications Principal Components

In the plot we see that the first loading vector places approximately equal weight on Assault, Murder and Rape with much less weight on UrbanPop.

Hence this component roughly corresponds to a measure of overall rates of serious crimes.

The second loading vector places most of its weight on UrbanPop and much less weight on the other three features.

Hence this component roughly corresponds to the level of urbanization of the state.

The crime-related variables (Murder, Assault, Rape) are located close to each other and the UrbanPop variable is far from the others.

This indicates that the crime-related variables are correlated with each other (states with high murder rates tend to have high assault and rape rates) and that the UrbanPop variable is less correlated with the others.



## **Applications Principal Components**

The loading vectors suggests that states with large positive scores on the first component (e.g. California, Nevada, Florida) have high crime rates and while states with negative scores on the first component (e.g. North Dakota) have low crime rates.

California also has a high score on the second component indicating a high level of urbanization while the opposite is true for states like Mississippi.

States close to zero on both components (e.g. Indiana) have approximately average levels of both crime and urbanization.